MovieLens Dataset Results

(a) RS domain

Approach	@k	NDCG	Precision	Recall
RS models				
LightFM	5	0.6467†	0.7646†	0.2719
	10	0.6481†	0.6887†	0.4774
	20	0.6701†	0.5273	0.6937
	50	0.7471	0.2643	0.8491
DeepFM	5	0.5490†	0.6375†	0.2450
	10	0.5738†	0.5970†	0.4459
	20	0.6326†	0.4922†	0.6955
	50	0.7354†	0.2684†	0.9068
BERT4Rec	5	**0.6753**	**0.7969**	0.2890
	10	**0.6955**	**0.7426**	0.5253
	20	**0.7441**	**0.5957**	0.7937
	50	**0.8297**	**0.2992**	0.9654
LightGCN	5	0.6080	0.6932	**0.2919**
	10	0.6330	0.6306	0.5096
	20	0.6909	0.5046	0.7618
	50	0.7871	0.2679	0.9631
ItemKNN	5	0.4375	0.5018	0.2602
	10	0.5420	0.5311	**0.5257**
	20	0.6422	0.4582	**0.8062**
	50	0.7159	0.2368	**0.9702**
IR → RS mapping				
Centroid Vector^a	5	0.3786	0.3654	0.1563
	10	0.3738	0.3259	0.2682
	20	0.4013	0.2766	0.4363
	50	0.5564	0.2018	0.7593
Tag Query^b	5	0.5413	0.5209	0.2157
	10	0.5071	0.4336	0.3440
	20	0.4790	0.3080	0.4672
	50	0.5475	0.1643	0.6054

(b) IR domain

Approaches	@k	NDCG	Precision	Recall
IR models				
BM25	5	0.0641	0.0597	0.0066
	10	0.0555	0.0460	0.0099
	20	0.0463	0.0352	0.0142
	50	0.0390	0.0246	0.0202
Dense Retriever	5	0.0697	0.0692	0.0021
	10	0.0734	0.0731	0.0042
	20	0.0745	0.0762	0.0082
	50	0.0770	**0.0772**	0.0198
Bi-Encoders	5	0.0845	0.0780	0.0097
	10	0.0771	0.0696	0.0160
	20	0.0700	0.0602	0.0255
	50	0.0671	0.0498†	0.0464
KNN-SPLADE	5	0.0597	0.0551	0.0085
	10	0.0541	0.0482	0.0138
	20	0.0524	0.0454	0.0224
	50	0.0513	0.0360	0.0398
ColBERT	5	**0.1248**	**0.1123**	0.0142
	10	**0.1183**	**0.1033**	0.0210
	20	**0.1093**	**0.0913**	0.0351
	50	0.0983	0.0753	0.0610
RS → IR mapping				
Query-as-User^c	5	0.0834	0.0806	**0.0265**
	10	0.0828	0.0761	0.0392†
	20	0.0845	0.0706	0.0609†
	50	0.1027	0.0617†	0.1037†
Retrieval-as-User^c	5	0.0847	0.0819	0.0232
	10	0.0813	0.0763	**0.0428**
	20	0.0778	0.0640	**0.0723**
	50	**0.1040**	0.0452†	**0.1299**

Table 1: MovieLens: BR2IDGE top-k mean results. **Bold** marks best values per @k/metric; † indicates statistical equivalence (t-test, p-mode=holm, $\alpha = 0.05$). Hybrid backbones: ^{a} Centroid Vector uses *Bi-Encoder*; ^{b} Tag Query uses *BM25*; ^{c} RS → IR strategies use *LightFM*.